

## Unit 404 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Cable**, 51.2 ac, BAF 20 variable-radius cruise on a 300 ft grid, 27 plots, 281 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

**Table 1. Stand summary from the BAF 20 cruise, per acre**

| STAND DENSITY        |                                    | STAND INFO          |                                                |
|----------------------|------------------------------------|---------------------|------------------------------------------------|
| Basal area           | 208 sq ft/ac (SE 3.7 %)            | Type                | Sierran mixed conifer                          |
| Trees per acre       | 61                                 | CWHR size / density | 5 / D (cover 87 %)                             |
| QMD                  | 25.0 in                            | SDI                 | 252 (43 % of max 580)                          |
| SAWTIMBER (>= 10 in) |                                    | BIOMASS (< 10 in)   |                                                |
| BA   QMD             | 207 sq ft/ac   25.7 in             | BA   QMD            | 1 sq ft/ac   5.9 in                            |
| TPA                  | 57                                 | TPA                 | 4                                              |
| Volume               | 39.2 MBF/ac (7803 cu ft)           | Volume              | 0.2 green tons/ac (10 cu ft)                   |
| TREATMENT TARGET     |                                    | CRUISE DESIGN       |                                                |
| Leave                | 35 % of max SDI = ~168 sq ft/ac BA | Sampling error      | 7.6 % at 95 % (t = 2.06, n = 27)               |
| Species (BA)         | WF 67, PP 64, DF 30, IC 27, SP 20  | Stratum standard    | 40 % - MEETS; 27 plots vs 20-plot practice: ok |

Cubic volume = total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium) net of defect; Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006); green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1). Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables exclude records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: [cruise\\_data.xlsx](#), Standards and assumptions.

**Table 2. Stand table by 4-inch DBH class, per acre**

| DBH class, in | Trees      | TPA         | BA           | cu ft       | MBF         |
|---------------|------------|-------------|--------------|-------------|-------------|
| 4-7.9         | 1          | 3.9         | 0.7          | 10          | 0.1         |
| 8-11.9        | 4          | 4.4         | 3.0          | 50          | 0.3         |
| 12-15.9       | 10         | 6.5         | 7.4          | 181         | 0.9         |
| 16-19.9       | 27         | 11.5        | 20.0         | 555         | 2.8         |
| 20-23.9       | 36         | 10.2        | 26.7         | 802         | 4.0         |
| 24-27.9       | 43         | 8.9         | 31.9         | 1062        | 5.3         |
| 28-31.9       | 38         | 5.8         | 28.1         | 1080        | 5.4         |
| 32-35.9       | 37         | 4.3         | 27.4         | 1059        | 5.3         |
| 36-39.9       | 22         | 2.1         | 16.3         | 699         | 3.5         |
| 40-43.9       | 13         | 1.0         | 9.6          | 411         | 2.1         |
| 44-47.9       | 17         | 1.1         | 12.6         | 599         | 3.0         |
| 48-51.9       | 17         | 0.9         | 12.6         | 654         | 3.3         |
| 52-55.9       | 3          | 0.1         | 2.2          | 115         | 0.6         |
| 56-59.9       | 3          | 0.1         | 2.2          | 110         | 0.6         |
| 60-63.9       | 10         | 0.4         | 7.4          | 425         | 2.1         |
| <b>All</b>    | <b>281</b> | <b>61.3</b> | <b>208.1</b> | <b>7814</b> | <b>39.2</b> |

**Table 3. Stock table by species, per acre**

| Species    | Trees      | TPA         | BA           | QMD         | cu ft       | MBF         |
|------------|------------|-------------|--------------|-------------|-------------|-------------|
| WF         | 90         | 23.7        | 66.7         | 22.7        | 2425        | 12.2        |
| PP         | 87         | 16.6        | 64.4         | 26.6        | 2704        | 13.6        |
| DF         | 41         | 7.5         | 30.4         | 27.2        | 1099        | 5.5         |
| IC         | 36         | 6.7         | 26.7         | 27.0        | 792         | 4.0         |
| SP         | 27         | 6.7         | 20.0         | 23.4        | 793         | 4.0         |
| <b>All</b> | <b>281</b> | <b>61.3</b> | <b>208.1</b> | <b>25.0</b> | <b>7814</b> | <b>39.2</b> |

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 404: plots on the 300 ft grid (orange = QA flag)

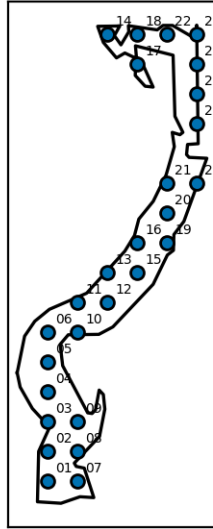


Figure 1. Plot locations with QA status.

**QA findings:** no issues found; unit accepted

**Table 4. Plot list**

| Plot   | Trees in | BA sq ft/ac | Slope % | Status |
|--------|----------|-------------|---------|--------|
| 404-01 | 11       | 220         | 65      | ok     |
| 404-02 | 11       | 220         | 50      | ok     |
| 404-03 | 13       | 260         | 39      | ok     |
| 404-04 | 10       | 200         | 67      | ok     |
| 404-05 | 11       | 220         | 57      | ok     |
| 404-06 | 11       | 220         | 56      | ok     |
| 404-07 | 12       | 240         | 54      | ok     |
| 404-08 | 9        | 180         | 48      | ok     |
| 404-09 | 13       | 260         | 60      | ok     |
| 404-10 | 6        | 120         | 41      | ok     |
| 404-11 | 12       | 240         | 52      | ok     |
| 404-12 | 11       | 220         | 60      | ok     |
| 404-13 | 12       | 240         | 44      | ok     |
| 404-14 | 9        | 180         | 48      | ok     |
| 404-15 | 8        | 160         | 59      | ok     |
| 404-16 | 10       | 200         | 58      | ok     |
| 404-17 | 11       | 220         | 36      | ok     |
| 404-18 | 10       | 200         | 47      | ok     |
| 404-19 | 4        | 80          | 46      | ok     |
| 404-20 | 10       | 200         | 71      | ok     |
| 404-21 | 11       | 220         | 57      | ok     |
| 404-22 | 10       | 200         | 43      | ok     |
| 404-23 | 10       | 200         | 40      | ok     |
| 404-24 | 10       | 200         | 57      | ok     |
| 404-25 | 11       | 220         | 50      | ok     |
| 404-26 | 12       | 240         | 50      | ok     |
| 404-27 | 13       | 260         | 36      | ok     |